

Spontaneous Regression of Incidental Meningioma in the Posterior Cranial Fossa: A Case Report and Literature Review

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We report a rare case of spontaneous regression of a meningioma located in the posterior cranial fossa. A 60-year-old female patient presented with a headache. Both physical and neurological examinations were normal. Her medical history included type 2 diabetes mellitus. MRI showed a homogeneously enhancing mass in the left cerebellopontine angle. The volume of the tumor was 1.673 cm³. Radiological features, including the dural tail sign and the cerebrospinal fluid cleft sign, suggested the presence of a meningioma. Given these characteristics, we recommended serial follow-up MRI examinations. Unfortunately, the patient was lost to follow-up. Ten years later, she returned with a headache to reevaluate the previously diagnosed tumor. The follow-up MRI revealed significant regression and increased intratumoral calcification, with the tumor now measuring only 0.256 cm³. This report illustrates that meningiomas diagnosed incidentally can undergo spontaneous regression, a phenomenon that clinicians should be aware of.

Keywords

Meningioma; Posterior cranial fossa; Spontaneous regression; Conservative management.

INTRODUCTION

Meningiomas are among the most common primary central nervous system tumors [1]. A significant number of these tumors are asymptomatic [2]. For patients with asymptomatic meningiomas, serial follow-up is advised, and surgical intervention is recommended if the tumor shows considerable growth or if symptoms develop.

Most meningiomas remain stable or increase in size during the follow-up period [2-4]. Contrastingly, spontaneous regression of meningiomas is rare. This phenomenon has been observed in cases associated with intratumoral hemorrhage, infarction, or hormonal influences [5]. In this report, we present a rare case of spontaneous regression of a meningioma in the posterior cranial fossa.

CASE REPORTS

A 60-year-old female patient presented with a headache. Both her physical and neurological examinations were normal. Her medical history included type 2 diabetes mellitus (DM), diagnosed 13 years prior, which was treated with metformin hydrochloride (1,000 mg/day). The patient's hemoglobin A1c (HbA1c) level was 6.8%. MRI revealed a round mass in the left cerebellopontine angle (Fig. 1A-C). Axial T2-weighted MRI displayed hyperintensity along with the cerebrospinal fluid cleft sign (Fig. 1A), while axial T1-weighted MRI (Fig. 1B) showed isointensity. The axial T1-weighted gadolinium-enhanced MRI indicated a homogeneously enhanced mass featuring the dural tail sign (Fig. 1C). Notably, there was no peritumoral edema (Fig. 1A). The tumor measured 1.673 cm³. Based on these radiological characteristics, the tumor was presumed to be a meningioma. No neurological symptoms or signs associated with the meningioma were present, leading to a diagnosis of an incidental meningioma in the posterior cranial fossa. Given the tumor's characteristics and the patient's age, we recommended observation with a follow-up MRI scheduled for 1 year later. However, the patient was lost to follow-up.

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After 10 years, the patient returned, experiencing a primary headache and seeking information about the previously diagnosed tumor. Follow-up MRI revealed a significant reduction in tumor volume. The tumor volume was now 0.256 cm^3 , representing an 85% reduction from its initial volume. The contrast enhancement pattern appeared similar to that seen in the previous images; however, a low-signal lesion, suspected to represent central calcification, was observed within the tumor (Fig. 1D-F). However, the patient's diabetes status had progressed slightly. The patient's HbA1c level was 7.9% and DM was managed with gliclazide (60 mg/day) and vildagliptin/metformin hydrochloride (100/1,000 mg/day). Furthermore, comorbid chronic kidney disease (stage III) was noted.

DISCUSSION

Meningiomas are among the most common intracranial tumors, and a significant number are managed with a "wait and

see" approach. These tumors are generally considered slow-growing [2]. The majority of meningiomas either remain static or increase in size [2,3]. Spontaneous regression of meningiomas, however, is rare. In this report, we have documented a case of spontaneous regression of a meningioma in the posterior cranial fossa. The tumor was relatively small, prompting us to recommend serial follow-up MRI examinations. Unfortunately, the patient was lost to follow-up. After 10 years, the patient returned for evaluation, which made it impossible to assess any changes in tumor size during the interim. Nevertheless, the absence of specific symptoms during this period suggests that the tumor's condition remained stable. The underlying cause of our patient's spontaneous tumor shrinkage remains uncertain, but we speculate that DM may have played a role in this case [6,7]. During the follow-up period, the patient's diabetes status gradually worsened and was further complicated by diabetic nephropathy. Although the precise underlying mechanism in the present case remains uncertain, the

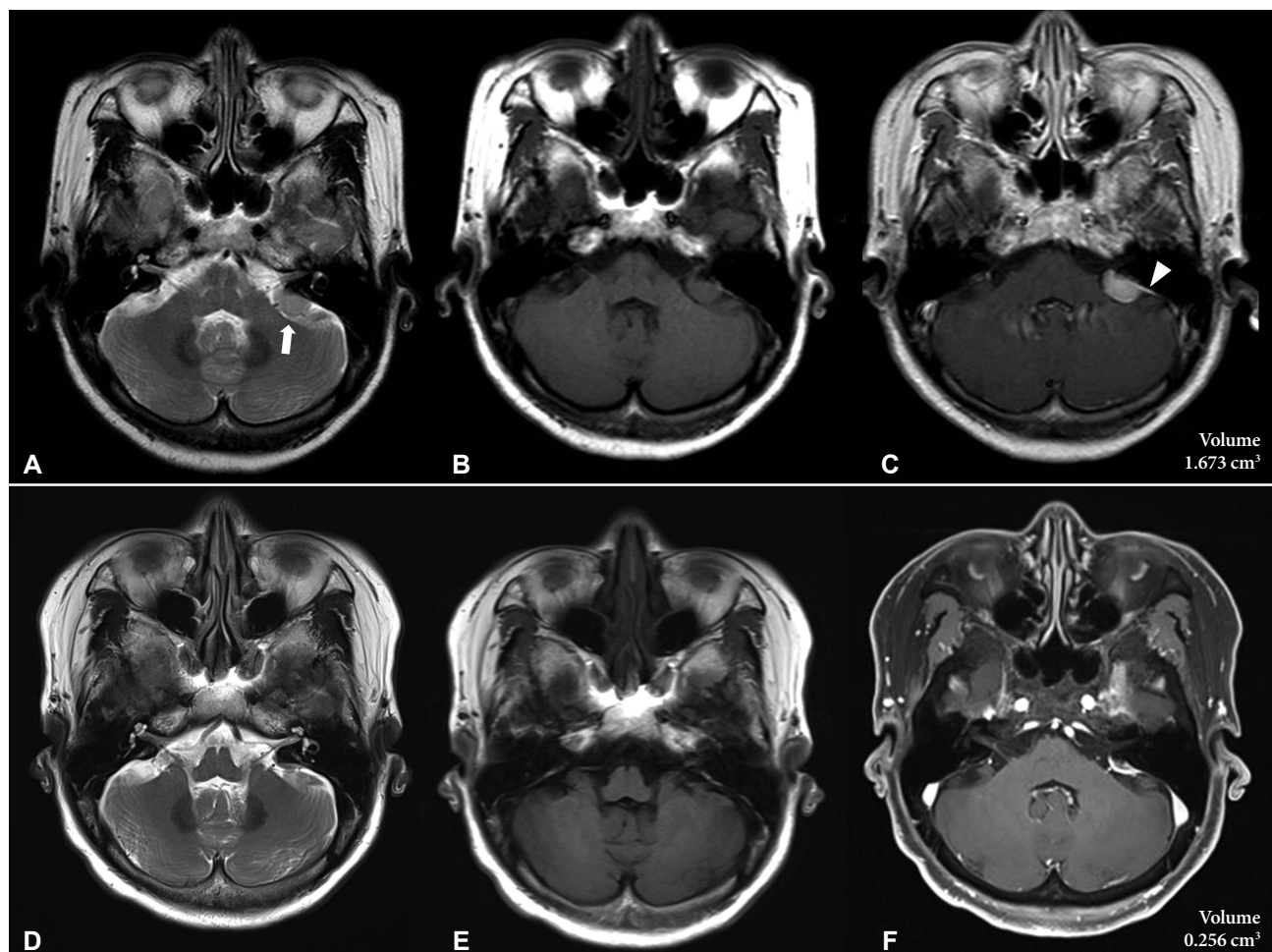


Fig. 1. MRI images of the patient. A-C: The lesion was slightly hyperintense on axial T2-weighted MRI (A) and isointense on axial T1-weighted MRI (B). A cerebrospinal fluid cleft sign (arrow) was seen on axial T2-weighted MRI (A). Axial T1-weighted gadolinium-enhanced MRI (C) showed a well-enhancing round mass with the dural tail sign (arrowhead). The volume of the tumor was 1.673 cm^3 . D-F: Follow-up MRI after 10 years showed a significant reduction, with intratumoral calcification. The volume of the tumor was 0.256 cm^3 .

exacerbation of DM may be a plausible contributing factor to the observed tumor regression. This mechanism has been suggested in several studies. Hirota et al. [6] proposed that meningioma regression may result from calcification of tumor cells facilitated by elevated glucose levels. Moreover, Sinha and Vyavahare [8] demonstrated that medial arterial calcification is most frequently observed in the distal arteries of individuals with diabetes mellitus, end-stage renal disease, or advanced age, and that glucose plays a significant role in enhancing osteogenesis mediated by elastin peptides and transforming growth factor-beta 1.

The natural history of meningiomas has been well-documented. Nakamura et al. [3] studied the growth rates and tu-

mor volume doubling times of incidental meningiomas, finding that the majority (66%) of growth rates were less than 1 cm³/year and the mean tumor doubling time was 21.6 years. Their findings indicated that tumor growth is associated with patient age, but initial tumor size is not a predictive factor for tumor growth [3]. Behbahani et al. [4] conducted a prospective study on the natural history of incidental meningiomas involving 64 cases, showing that 75% of tumors increased (>15%), 20.3% remained unchanged, and 4.7% decreased (>15%) in volume. More recently, Nakasu S and Nakasu Y [2] analyzed the natural history of meningiomas through meta-analyses, reporting that two-thirds of meningiomas showed radiological progression and that growth rates typically plateaued at

Table 1. Summary of reported cases of meningioma showing spontaneous regression

Author (year)	Age (yr)/ Sex	Location	Initial volume (cm ³)	Final volume (cm ³)	Regression rate (%)	Follow-up duration (month)	Possible factors
Pozzati et al. [9] (2007)	59/F	Multiple	NA	NA	Partial regression	24	Discontinuation of progesterone
Shimizu et al. [10] (2008)	80/M	Frontal base	50 mm	NA	Remarkable regression	33	Discontinuation of progesterone
De Almeida et al. [11] (2009)	66/F	Parietal	5.5	NA	90	12	Intratumoral hemorrhage
Vadivelu et al. [13] (2010)	63/F	Multiple	0.6–49.35	0–1.49	75–100	60	Discontinuation of progesterone
Gonçalves et al. [12] (2010)	46/F	Olfactory groove	16.0	1.9	88	6	Discontinuation of cyproterone
Hirota et al. [6] (2014)	66/F	Falx	25.5	9.9	61.2	84	Unknown
Kerschbaumer et al. [16] (2016)	32/F	Sphenoid ridge	53.1	27.6	48	2	Hormonal factor after pregnancy
Witherick et al. [17] (2016)	75/F	Falx	53 mm	NA	Partial regression	84	Unknown
Galloway et al. [14] (2016)	73/F	Pineal region	40 mm	18 mm	55	36	Central necrosis
Hoegestoel et al. [15] (2016)	59/M	Sphenoid ridge	29.9	6.7	78	37	Alpha-1 adrenoceptor agonist
Galloway et al. [26] (2017)	56/F	Falx	25 mm	18 mm	28	120	Interferon beta-1a
Kalamarides et al. [18] (2017)	26/F	Parietal/Falx	68/1.6	13/0.3	81/80	12	Discontinuation of progesterone
	43/F	Multiple	2.3–64	1–11	17–83	12	
Passeri et al. [19] (2019)	37/F	Both sphenoid	54.5/1.8	13.1/NA	76/NA	12	Discontinuation of NOMAC
	68/F	Clivus	12.4	10.3	16.9	4	
	54/F	Frontal/Clinoid	5.7/9.4	5.2/7.9	9/16	7	
Kumaria et al. [7] (2020)	58/F	Sphenoid	36	11.2	68.9	84	Unknown
Shahin et al. [20] (2021)	40/F	Multiple	10–11 mm	7–8 mm	NA	4	Discontinuation of progesterone
Takada et al. [22] (2022)	68/F	Posterior fossa	2.3	0.5	78.3	144	Unknown
Kim et al. [21] (2022)	73/F	Sphenoid	58.6	37.2	36.6	120	Vascular
Bailey et al. [23] (2024)	NA/F	Multiple	0.3–14.9	0.1–3.2	67–79	60	Discontinuation of progesterone
Noda et al. [5] (2024)	66/M	Petroclival	2.5	1.0	62.5	156	Unknown
Present case	60/F	Posterior fossa	1.7	0.3	85	120	Unknown

NA, not available; NOMAC, nomegestrol acetate.

4–5 years. They also found that factors such as younger age, absence of calcification, and high intensity on T2-weighted MRI were associated with radiological progression and rapid growth, but not with symptomatic progression [2].

Conversely, spontaneous shrinkage of meningiomas is rare. A literature review was conducted using the PubMed database with the search terms “meningioma” and “spontaneous regression.” To the best of our knowledge, there have been a total of 20 reports, accounting for 23 cases of spontaneous meningioma regression reported in English to date; all 23 cases, including the current case, are listed in Table 1 [5-7,9-23]. The average age at diagnosis was 56.9 years (range: 26–80 years). The average observation period was 54.3 months (range: 2–156 months). The reported reduction in tumor volume ranged from 9% to 100%. Most of the cases involved female patients. Hormonal-related conditions were the most commonly reported causes of spontaneous regression. As shown in Table 1, several case reports have suggested a potential relationship between the discontinuation of hormone therapy, particularly progesterone, and meningioma regression. While the exact mechanism underlying this relationship remains unclear, it may be related to the expression of hormone receptors—particularly the progesterone receptor—in meningiomas [24,25]. Other reported causes of natural tumor regression include factors related to vascular changes and hemorrhage. The follow-up period for hormone-related regressions was shorter than for other causes [9,10,12,13,15,16,18-20,23,26]. In our case, the patient was a postmenopausal woman at the time of diagnosis and had only used medications related to diabetes mellitus.

A limitation of this report is the absence of interim MRI scans, which obscures any understanding of the changes that may have occurred over time. The exact mechanism behind the tumor’s spontaneous regression remains unclear. As mentioned earlier, considering the patient’s history of diabetes mellitus, we can only speculate about a possible association. However, the lack of reported symptoms during this period suggests that the tumor remained stable. Further research is necessary to elucidate the underlying causes of spontaneous regression in meningiomas.

In conclusion, we present a relatively rare case of meningioma exhibiting spontaneous regression. Understanding the natural course of meningioma is crucial for optimizing treatment strategies and follow-up plans. This case may offer valuable insights into progression patterns. Moreover, this case report suggests that observation can be an appropriate treatment option for cases where the patient is an older female and the initially detected meningioma is relatively small.

Ethics Statement

The Institutional Review Board (IRB) exempted informed consent due to its retrospective nature and minimal risk of harm to the patient. This report was conducted according to the guidelines of the Declaration of Helsinki for biomedical research (IRB No. 2022-08-029).

Availability of Data and Material

The datasets generated or analyzed during the study are available from the corresponding author upon reasonable request.

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Conflicts of Interest

Youngbeom Seo, a contributing editor of the *Brain Tumor Research and Treatment*, was not involved in the editorial evaluation or decision to publish this article. All remaining authors have declared no conflicts of interest.

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